

Europa Subsurface Ocean & Ice Shell Thermal Architecture (Squyres et al. 1983, Cassen et al. 1979, 1980)

SPACE VACUUM (Surface $T_s \approx 100$ K, Radiation $F_{\text{rad}} = \epsilon \sigma T_s^4$)

Surface (0 km)

Cryovolcanic Venting

COLD BRITTLE ICE LITHOSPHERE (0-8 km)

Conductive Heat Transport: $q(z) = -k(T) dT/dz$

Cycloid Fractures & Flexural Cracking ($\sigma_{\text{tide}} > 40$ kPa)

Brittle Transition (8 km)

WARM DUCTILE ICE ASTHENOSPHERE (8-20 km)

Viscoelastic Tidal Dissipation Peak: $\dot{\epsilon}_{\text{tide}} = \frac{21}{2} \frac{\rho n^5 R^2 e^2}{\mu Q}$

Subsolidus Thermal Convection & Diapirism ($T \sim 220 - 271$ K)

Basal Melting Boundary $z = H_{\text{eq}} \approx 20$ km ($T_b = 271.4$ K, $P_{\text{base}} = 24.1$ MPa)

Base of Ice (20 km)

Total Heat Flux

$F_{\text{total}} \approx 30$ mW/m²

→ $H_{\text{eq}} \approx 19 - 20$ km

GLOBAL SUBSURFACE LIQUID H₂O OCEAN (20-120 km)
Decouples Ice Shell from Rocky Mantle (Amplifies Tidal Flexing)
Hydrothermal Circulation & Habitability Environment

Hydrothermal Plumes

Warm Upwelling

Cold Downwelling

Seafloor (120 km)

SILICATE ROCKY MANTLE & METALLIC CORE (>120 km)

Radiogenic Heating: $Q_{\text{radio}} \approx 200$ GW ($F_{\text{core}} \approx 6.5$ mW/m²)

Hydrothermal Vents & Seafloor Tidal Friction